

Notes: Chevalier & Ellison (JPE, 1997)

Risk Taking by Mutual Funds as a Response to Incentives

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1 Big picture

- **Question.** Do mutual funds change the riskiness of their portfolios in response to the incentives created by investors' flow-performance relationship?
- **Agency problem.** Investors want funds to maximize risk-adjusted returns, but fund companies earn fees from assets under management. Therefore, the fund company may prefer portfolio choices that maximize expected future inflows rather than investor welfare.
- **Key mechanism.** If investor flows respond nonlinearly to year-end performance, then a manager's incentive to increase or decrease risk depends on the fund's year-to-date position.
- **Empirical design.** The authors first estimate the flow-performance relation semi-parametrically. Then they transform the estimated relation into an incentive to take risk. Finally, they ask whether funds alter September-to-December portfolio risk in the predicted direction.
- **Main finding.** Funds do alter risk in response to incentives. The strongest evidence is for young and small funds, whose future flows are most sensitive to performance.
- **Most intuitive case.** A young fund that is moderately behind the market has an incentive to gamble: losing more does not worsen expected flows much, while catching up sharply improves expected flows.
- **Less intuitive case.** A fund that is moderately ahead may have an incentive to reduce risk and lock in gains, while a fund far ahead may again have an incentive to gamble because the upside of becoming a star fund is large.
- **Economic takeaway.** The paper treats investor flows as an implicit incentive contract. Because the contract is nonlinear, it can induce fund companies to change risk for business reasons, not because doing so maximizes investors' risk-adjusted returns.

2 Incremental contribution of the paper

- **From performance-flow facts to incentive design.** Earlier work documented that mutual fund flows respond to performance. This paper's incremental step is to interpret that relationship as an implicit incentive scheme facing asset managers.
- **Nonlinear incentives, not just tournament ranking.** Brown, Harlow, and Starks (1996) and related tournament-style work ask whether midyear losers gamble. Chevalier and Ellison (1997) derive the entire risk-taking incentive from the estimated flow curve, allowing the incentive to vary continuously with year-to-date performance.
- **Semi-parametric flow estimation.** Instead of imposing a two- or three-segment flow-performance function, the paper estimates the shape flexibly. This matters because risk incentives depend on curvature, not merely on average flow-performance sensitivity.
- **Direct portfolio evidence.** The paper uses holdings data from September and December to measure actual changes in portfolio risk. This is stronger than relying only on return-based risk measures because it observes changes in the equity portfolio itself.
- **Agency interpretation of mutual fund behavior.** The paper connects household/investor demand, mutual fund company value, and end-of-year portfolio risk choices. It shows that even if investors respond rationally to performance, their aggregate flow behavior can create distorted managerial incentives.
- **Reusable methodology.** The method can be applied to other asset-management behaviors: window dressing, herding, style drift, or any setting where an empirically estimated payoff schedule can be translated into managerial incentives.

3 Data and measurement

3.1 Flow-performance sample

- The primary data source is Morningstar’s January 1994 *Mutual Funds OnDisc*.
- The flow sample contains growth and growth-and-income mutual funds observed during years between 1983 and 1993.
- The flow regression sample has 449 funds and 3,036 fund-year observations.
- The authors use Morningstar information on returns, assets under management, minimum initial purchase requirements, expense ratios, and merger history.
- Because *OnDisc* only includes funds operating as of January 1994, the authors use Morningstar’s deleted-fund list since 1989 to reconstruct data for funds dropped from the database after 1989.
- The authors exclude closed funds, primarily institutional funds, funds with initial purchase requirements above \$25,000, high-expense-ratio funds, funds involved in mergers in the next year, funds less than two years old, and funds with less than \$10 million in assets.

Interpretation: The final sample is designed to study retail mutual fund flow incentives. The exclusions remove cases where proportional flow growth is likely noisy or economically different from normal retail fund demand.

3.2 Flow definition and performance variables

The main dependent variable is next-year proportional flow, net of internal asset growth:

$$Flow_{i,t+1} = \frac{Assets_{i,t+1} - Assets_{i,t}}{Assets_{i,t}} - r_{i,t+1}. \quad (1)$$

- $r_{it} - r_{mt}$ is year- t fund return minus the value-weighted market return.
- Lagged excess returns $r_{i,t-1} - r_{m,t-1}$ and $r_{i,t-2} - r_{m,t-2}$ control for delayed investor response.
- The year- $t+1$ excess return is included because investors may respond to within-year returns when making contributions and redemptions.
- Industry growth controls for aggregate growth in equity mutual funds.
- Fund size is measured as log assets relative to the geometric mean of the sample.

Interpretation: The flow variable is a business-value measure from the fund company’s perspective. Since management fees are proportional to assets, expected flow growth is a proxy for the future value of the fund franchise.

3.3 Age categories

- Funds are classified into age categories: 2, 3, 4, 5, 6–7, 8–10, and 11 or more years old.
- The paper estimates young funds, ages 2–5, separately from old funds, ages 6 or more.
- The omitted categories are age 2 for young funds and age 11 or more for old funds.

Interpretation: Young funds should have more performance-sensitive flows because investors have less history with which to infer quality. This is central: if flows are more sensitive for young funds, the business incentive to manipulate risk should be stronger for them.

3.4 Portfolio holdings sample

- The risk-change analysis uses a Morningstar holdings data set with complete equity portfolios.
- The key sample contains 839 fund-years for 398 growth and growth-and-income funds with both September and December portfolios.
- Each holding is matched to CRSP securities using names, tickers when available, and price consistency.
- The authors match 92.5% of security records, and risk calculations typically use about 89% of records after requiring enough historical return data.
- The main analysis generally restricts to funds for which at least 85% of the September portfolio value can be matched.

Interpretation: The holdings data allow the paper to measure actual risk changes from observed portfolio composition, not just from realized returns. This is important because realized returns can reflect market shocks rather than active changes in risk.

3.5 Risk measures

The paper studies several risk changes from September to December:

$$\Delta SD(r - r_m) = SD_{Dec}(r - r_m) - SD_{Sept}(r - r_m), \quad (2)$$

$$\Delta SD(r - \beta r_m) = SD_{Dec}(r - \beta r_m) - SD_{Sept}(r - \beta r_m), \quad (3)$$

$$\Delta |\beta - 1| = |\beta_{Dec} - 1| - |\beta_{Sept} - 1|. \quad (4)$$

- $SD(r - r_m)$ captures volatility relative to the market return.
- $SD(r - \beta r_m)$ focuses more on idiosyncratic residual risk after market beta adjustment.
- $|\beta - 1|$ captures deviation from market beta exposure.
- Later, the paper checks total portfolio risk changes using time-series return-based variance measures.

Interpretation: The main incentive being studied is not simply “take more beta.” It is a broader incentive to move away from a market-index-like portfolio when the flow curve makes upside more valuable than downside costly.

4 Models and empirical specifications

4.1 Semi-parametric flow-performance model

The estimated flow model is:

$$Flow_{i,t+1} = \sum_k \gamma_k Age_{k,it} f(r_{it} - r_{mt}) + \sum_k \delta_k Age_{k,it} \quad (5)$$

$$+ \alpha_1(r_{i,t-1} - r_{m,t-1}) + \alpha_2(r_{i,t-2} - r_{m,t-2}) \quad (6)$$

$$+ \alpha_3(r_{i,t+1} - r_{m,t+1}) + \alpha_4 IndustryGrowth_{t+1} + \alpha_5 \log(Assets_{it}) + e_{i,t+1}. \quad (7)$$

Interpretation: The unknown function $f(\cdot)$ is the central object. Its slope and curvature determine how valuable an additional point of year-end performance is to the fund company.

4.2 Three-step semi-parametric estimation

1. Estimate the linear controls α using Robinson’s partialling-out procedure: kernel-regress both flows and controls on year- t excess returns, then regress residualized flows on residualized controls.
2. Estimate age-specific scale and shift terms γ_k and δ_k by comparing age-specific nonparametric functions.
3. Estimate the normalized function f by kernel-regressing residualized and rescaled flows on year- t excess returns.

Interpretation: This structure lets the authors flexibly estimate the flow-performance curve while still controlling for lagged returns, fund size, and industry growth. Flexibility matters because the derived risk incentives depend on nonlinearities in f .

4.3 Risk incentive measure

Let s denote the fund’s January–September excess return. The paper asks how expected future flow changes if the fund increases fourth-quarter risk. A generic representation is:

$$RiskIncentive_i(s) = E \left[\widehat{Flow}(s + q^{High}) \right] - E \left[\widehat{Flow}(s + q^{Low}) \right], \quad (8)$$

where q^{High} and q^{Low} represent higher- and lower-risk fourth-quarter return distributions.

Interpretation: A positive value means that raising risk increases expected future flows. The measure is shaped by the curvature of the flow-performance relationship: if upside outcomes generate large inflows and downside outcomes do not generate equally large outflows, risk taking is rewarded.

4.4 Portfolio risk-change regression

The main test uses September-to-December changes in portfolio risk:

$$\Delta Risk_i = \beta_0 + \beta_1 RiskIncentive_i + \beta_2 RiskIncentive_i \times \log(Assets_i) + \beta_3 \log(Assets_i) + \beta_4 SeptRisk_i + u_i. \quad (9)$$

Interpretation: The hypothesis is $\beta_1 > 0$, with $\beta_2 < 0$ if smaller funds respond more strongly. This specification links an independently estimated incentive schedule to actual portfolio changes.

4.5 Piecewise linear risk-change model

To compare actual risk changes with the shape of the predicted incentives, the paper estimates:

$$\Delta SD(r_i - r_m) = \alpha_0 + \alpha_1 \log(Assets_i) + \alpha_2 SeptSD_i + h(s_i) + \varepsilon_i, \quad (10)$$

where s_i is January–September excess return and $h(\cdot)$ is continuous and piecewise linear with two kink points and three slopes.

Interpretation: The piecewise model asks whether the visual shape of actual risk changes resembles the theoretical incentive curve: for young funds, the expected pattern is positive, negative, positive slopes around two turning points.

5 Identification strategy and empirical logic

5.1 Step 1: Estimate investor demand as an implicit incentive contract

The first step identifies how future assets under management respond to past performance. The authors treat the flow-performance relation as a reduced-form payoff schedule for the fund company.

Interpretation: This is not a randomized treatment. The identifying assumption is that the estimated flow curve captures the relevant business incentives that managers face when choosing fourth-quarter risk.

5.2 Step 2: Convert the flow curve into risk-taking incentives

Because risk changes the distribution of possible year-end returns, the estimated flow curve maps a fund's September position into an expected payoff from increasing risk.

Interpretation: The nonlinear shape of the flow curve is essential. A linear flow curve would not create state-dependent gambling incentives of the type studied here.

5.3 Step 3: Test whether observed portfolio changes follow incentives

The timing is central:

- The fund's January–September performance is known before the final-quarter portfolio choice.
- September and December holdings measure risk changes over the final quarter.
- Next-year flows are the payoff that makes the portfolio change valuable to the fund company.

Interpretation: This timing supports the incentive interpretation: the variable used to compute incentives is known before the measured risk adjustment.

5.4 Why young and small funds are informative

- Young funds have shorter track records, so investor flows respond more to recent performance.
- Small funds can change their portfolios more easily and may have stronger franchise-value incentives.
- The strongest predicted and observed effects appear for young and smaller funds.

Interpretation: Heterogeneity by age and size is a built-in falsification logic. If risk changes were unrelated to flow incentives, there would be less reason for the response to be strongest where the estimated flow-payoff curve is strongest.

5.5 Main threats and limitations

- The flow equation is reduced-form and may reflect unobserved fund quality, marketing, or investor attention.
- The holdings sample is voluntary and may be selected.
- The pre-1989 Morningstar data have survivorship-bias concerns, although the working-paper robustness checks reportedly produce similar results in less biased subsamples.

- Risk measures are constructed from equity holdings and may not capture all risk exposures.
- Managers may change risk for reasons unrelated to flow incentives, such as new information or style changes.

Interpretation: The evidence is strongest as a consistency test: estimated incentives and observed risk changes line up in timing, sign, and heterogeneity. The design does not prove that every individual risk change is caused by incentives.

6 Tables and figures

6.1 Table 1: Summary Statistics for Flow-Performance Regressions

Read:

- The flow sample has 3,036 fund-year observations: 780 young fund-years and 2,256 old fund-years.
- Mean next-year flow is 12.2% for the full sample, 25.1% for young funds, and 7.9% for old funds.
- Young funds are smaller on average, with mean log assets 4.254 versus 5.172 for old funds.
- Mean excess returns are close to zero, which is useful because the semiparametric curve is centered around typical market-adjusted performance.

Interpretation: Table 1 shows why age matters. Young funds grow more rapidly and are smaller, so their franchise value is more sensitive to investor attention and performance records.

Economic message: The descriptive statistics motivate the paper’s age split: young funds face steeper flow incentives and therefore stronger potential risk-taking distortions.

TABLE 1

SUMMARY STATISTICS FOR FLOW-PERFORMANCE REGRESSIONS

VARIABLE	FULL SAMPLE		YOUNG FUNDS		OLD FUNDS	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Flow _{<i>i+1</i>}	.122	.646	.251	.700	.079	.620
<i>r_i</i> − <i>rm_i</i>	−.005	.084	−.006	.085	−.005	.083
<i>r_{i−1}</i> − <i>rm_{i−1}</i>	−.002	.087	.0002	.091	−.003	.086
<i>r_{i−2}</i> − <i>rm_{i−2}</i>	−.001	.080	.001	.073	−.002	.083
<i>r_{i+1}</i> − <i>rm_{i+1}</i>	−.008	.077	−.006	.084	−.009	.075
IndustryGrowth _{<i>i+1</i>}	.283	.173	.272	.181	.286	.169
log(Assets)	4.936	1.427	4.254	1.173	5.172	1.431

	AGE CATEGORY						
	2	3	4	5	6–7	8–10	11+
Number of fund-years	221	199	189	171	238	168	1,850

Table 1: Summary Statistics for Flow-Performance Regressions

6.2 Figures 1–2: Flow-Performance Relationship for Young and Old Funds

Read:

- Figure 1 plots the estimated flow-performance curve for age-2 funds with 90% confidence bands.
- For young funds, flows are strongly increasing and nonlinear: severe losers have outflows, moderate underperformers face a flatter region, and extreme winners receive very large inflows.
- The paper notes that an age-2 fund matching the market is expected to grow by about 15% next year, while a fund beating the market by 10 percentage points is expected to grow by about 55%.
- Figure 2 shows old funds. The relation is flatter, implying weaker incentives from recent performance.

Interpretation: These figures are the foundation of the paper. The curvature, not merely the slope, determines whether risk taking is rewarded or punished at each performance level.

Economic message: Young funds have a more convex and performance-sensitive payoff schedule, so they are more likely to distort risk near year-end.

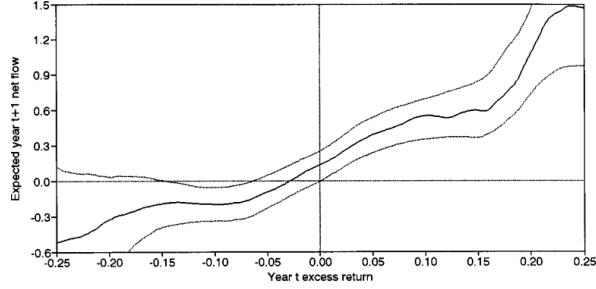


FIG. 1.—Flow-performance relationship $\hat{f}$ for young funds (age 2) with 90 percent confidence bands.

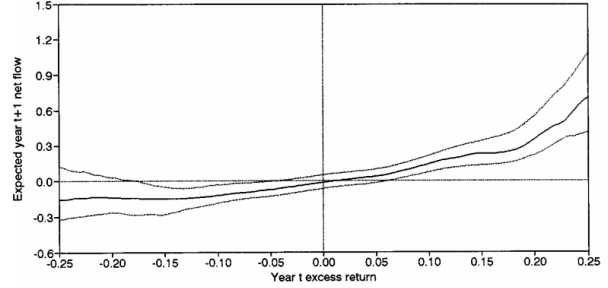


FIG. 2.—Flow-performance relationship $\hat{f}$ for old funds (age > 10) with 90 percent confidence bands.

(a) Figure 1: young funds, age 2

(b) Figure 2: old funds, age > 10

Flow-performance relationship with 90 percent confidence bands

6.3 Table 2: Coefficients from Semiparametric Flow-Performance Model

Read:

- Multiplicative coefficients for ages 3, 4, and 5 are negative relative to age 2, indicating that the flow curve is scaled down as young funds mature.
- Age-4 and age-5 funds have significantly lower flow sensitivity than age-2 funds.
- Lagged excess returns matter: for young funds, year $t - 1$ excess return has coefficient 1.86 and year $t - 2$ has coefficient 0.73.
- Year $t + 1$ excess return has a large coefficient, consistent with investors responding within the flow year and with internal asset growth effects.
- Industry growth has a nearly one-for-one effect on flows.

Interpretation: The table quantifies the age-dependent scaling of the nonparametric flow curve. It shows that the same performance outcome produces more flow response for younger funds than for mature funds.

Economic message: Investor learning is strongest early in a fund's life, making young managers especially exposed to flow-based incentives.

6.4 Figure 3: Risk Incentives with 90 Percent Confidence Bands

Read:

- Figure 3 transforms the flow-performance curves into incentives to increase portfolio risk at the end of September.
- For age-2 funds, the incentive is positive for funds somewhat behind the market and for funds far ahead of the market.
- The incentive can be negative for funds in the region where preserving the current outcome is more valuable than adding volatility.
- For old funds, the incentive curve is much weaker and less sharply patterned.

Interpretation: A manager somewhat behind the market can have a convex payoff: catching up generates substantial inflows, while falling further behind does not create equally large marginal outflows. This creates a gambling incentive.

TABLE 2
COEFFICIENTS FROM SEMIPARAMETRIC FLOW-PERFORMANCE
MODEL

INDEPENDENT VARIABLE	DEPENDENT VARIABLE: Flow _{t+1}	
	Young Funds (N = 780) (1)	Old Funds (N = 2,256) (2)
Multiplicative:		
Age 3 (γ_3)	-.28 (.25)	
Age 4 (γ_4)	-.34 (.20)	
Age 5 (γ_5)	-.46 (.22)	
Age 6-7 (γ_{67})		.30 (.79)
Age 8-10 (γ_{810})		.35 (.38)
Additive:		
Age 3 (δ_3)	.06 (.07)	
Age 4 (δ_4)	-.02 (.06)	
Age 5 (δ_5)	-.01 (.06)	
Age 6-7 (δ_{67})		.03 (.09)
Age 8-10 (δ_{810})		.02 (.04)
Excess return _{t-1}	1.86 (.25)	1.00 (.14)
Excess return _{t-2}	.73 (.31)	.29 (.16)
Excess return _{t+1}	2.49 (.26)	2.44 (.16)
Industry Growth _{t+1}	1.07 (.12)	.85 (.07)
log (Assets/ $\bar{A}$)	-.07 (.02)	-.04 (.01)
R ²	.38	.22

NOTE — Estimated standard errors are in parentheses

Table 2: Coefficients from Semiparametric Flow-Performance Model

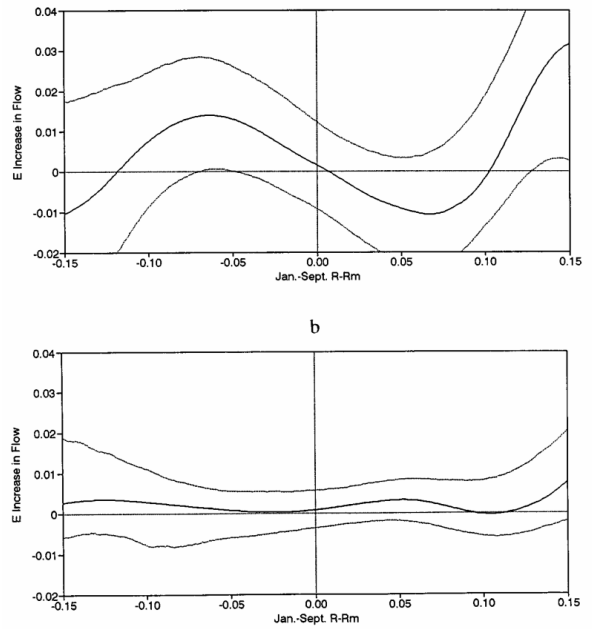


FIG. 3.—Risk incentives with 90 percent confidence bands. *a*, Funds of age 2. *b*,

Figure 3: Risk Incentives with 90 Percent Confidence Bands

Economic message: The implicit contract created by investors does not uniformly reward prudence. Depending on the fund's position, it can reward risk reduction or risk escalation.

6.5 Tables 3–4: Summary Statistics for Risk Change Regressions and September–December Risk Changes and Incentives

Read:

- Table 3 reports baseline risk-change variables. Mean September $SD(r-r_m)$ is 0.053 and mean $SD(r-\beta r_m)$ is 0.048.
- The average change in risk is small but positive: $\Delta SD(r-r_m) = 0.0019$ and $\Delta SD(r-\beta r_m) = 0.0017$.
- Table 4 tests whether risk changes respond to the calculated incentive.
- The coefficient on RiskIncentive is positive for both $\Delta SD(r-r_m)$ and $\Delta SD(r-\beta r_m)$.
- The interaction with log assets is negative, implying that the response declines with fund size.

Interpretation: Table 4 is the central statistical test. It links the predicted incentive from the flow curve to actual portfolio-risk changes. The negative interaction with size indicates that small funds are the main responders.

Economic message: The evidence supports an agency channel: funds do not merely face theoretical incentives; they appear to alter risk when the business payoff from doing so is higher.

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TABLE 3

Variable	Mean	Standard Deviation
Sept. SD($r - rm$)	.053	.022
Sept. SD($r - \beta rm$)	.048	.020
Sept. $ \beta - 1 $	.153	.115
Δ SD($r - rm$)	.0019	.008
Δ SD($r - \beta rm$)	.0017	.007
$\Delta \beta - 1 $	.006	.045
Risk Incentive	.0021	.003
log(Assets)	5.269	1.475
Sept. share matched	.936	
Dec. share matched	.918	

TABLE 4
SEPTEMBER–DECEMBER RISK CHANGES AND INCENTIVES ($N = 464$)

INDEPENDENT VARIABLE	DEPENDENT VARIABLE		
	$\Delta SD(r - rm)$ (1)	$\Delta SD(r - \beta rm)$ (2)	$\Delta \beta - 1 $ (3)
Risk Incentive	.87 (.28)	.73 (.24)	2.22 (1.40)
Risk Incentive - log(Assets)	-.15 (.05)	-.11 (.04)	-.57 (.28)
log(Assets)	.00013 (.0003)	-.00003 (.0002)	.003 (.002)
Sept. risk level	.05 (.02)	.03 (.02)	-.007 (.02)
Constant	-.001 (.002)	-.0002 (.002)	-.006 (.009)
R^2	.03	.03	.01

NOTE.—Estimated standard errors are in parentheses.

(a) Table 3: risk-change summary statistics

(b) Table 4: risk changes and incentives

Risk-change variables and main incentive regression

6.6 Table 5 and Figure 4: Actual Risk Changes as a Function of January–September Returns

Read:

- Table 5 estimates a piecewise linear model of actual equity risk changes as a function of January–September excess returns.
- For young funds, the first kink is estimated around -0.06 , matching the region where the risk incentive curve predicts a change in behavior.

- The young-fund slopes follow the predicted positive-negative-positive pattern, although not all estimates are precise.
- Figure 4 plots the fitted values from this model for young and old funds.
- The most visible result is that young funds moderately behind the market tend to raise equity portfolio risk.

Interpretation: The detailed shape evidence is noisier than the pooled incentive regression, but it provides a visual consistency check. The young-fund curve resembles the predicted incentive curve more than the old-fund curve does.

Economic message: Funds appear to gamble when the flow payoff to catching up is high, especially among young funds where reputational stakes are greatest.

TABLE 5
ACTUAL RISK CHANGES AS A FUNCTION OF JANUARY–
SEPTEMBER RETURNS

INDEPENDENT VARIABLE	DEPENDENT VARIABLE: $\Delta SD(r - rm)$	
	Young Funds ($N = 94$) (1)	Old Funds ($N = 375$) (2)
Constant	-.02 (.006)	-.003 (.002)
KINK1	-.06 (.02)	.03 (.03)
KINK2	.001 (.04)	.08 (.03)
SLOPELEFT	.09 (.04)	-.02 (.01)
SLOPEMID	-.07 (.07)	.06 (.08)
SLOPERIGHT	.003 (.02)	-.05 (.04)
log(Assets)	-.0007 (.0006)	.00002 (.0003)
Sept. $SD(r - rm)$	-.02 (.04)	.09 (.02)
R^2	.11	.08

NOTE.—Estimated standard errors are in parentheses.

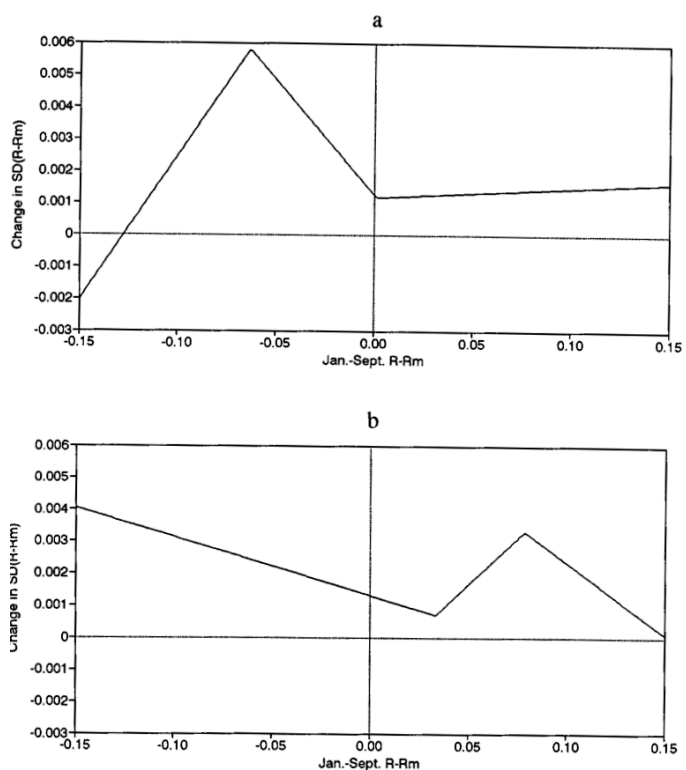


FIG. 4.—Pattern of actual risk changes: fitted values from piecewise linear model. *a*, Young funds' equity portfolios. *b*, Old funds' equity portfolios.

(a) Table 5: piecewise risk-change regression

(b) Figure 4: fitted risk changes from equity portfolios

Actual risk changes as a function of January–September returns

6.7 Table 6 and Figure 5: Risk Changes from Time Series as a Function of January–September Returns

Read:

- Table 6 repeats the piecewise analysis using time-series return-based risk changes instead of holdings-based equity risk changes.

- For young funds, the point estimates again show a positive-negative-positive slope pattern.
- For old funds, the pattern is more monotone: higher January–September returns are associated with larger risk increases.
- Figure 5 plots the fitted time-series risk-change curves.

Interpretation: The time-series results provide a broader but less clean risk measure. They confirm that risk changes are not purely an artifact of the matched equity holdings, but the precise shape is less tightly aligned with the incentive curve.

Economic message: The core result is robust enough to appear in return-based measures, but the cleanest evidence remains the holdings-based test for young and smaller funds.

TABLE 6
RISK CHANGES FROM TIME SERIES AS A FUNCTION OF
JANUARY–SEPTEMBER RETURNS

INDEPENDENT VARIABLE	DEPENDENT VARIABLE: $\Delta \text{TS var}(r - rm)$	
	Young Funds ($N = 869$) (1)	Old Funds ($N = 2,294$) (2)
Constant	.001 (.0008)	.001 (.0005)
KINK1	-.032 (.030)	-.036 (.032)
KINK2	.054 (.037)	.217 (.043)
SLOPELEFT	.024 (.010)	.024 (.006)
SLOPEMID	-.004 (.013)	.011 (.003)
SLOPERIGHT	.024 (.010)	.085 (.040)
R^2	.02	.03

NOTE.—Estimated standard errors are in parentheses.

(a) Table 6: time-series risk-change regression

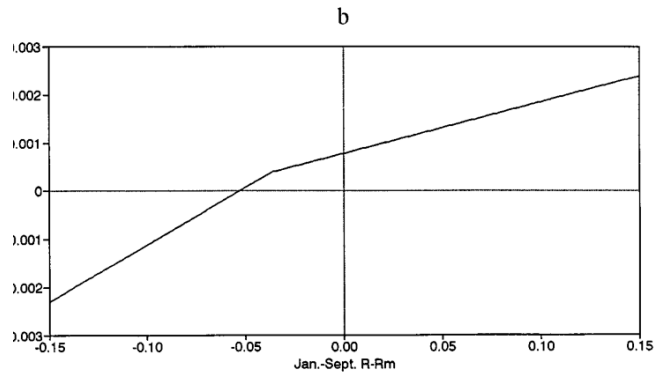
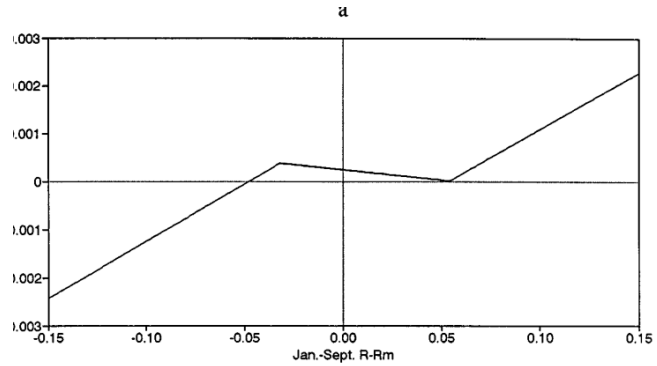


FIG. 5.—Pattern of actual risk changes via time series: fitted values from piecewise or model. *a*, Young funds. *b*, Old funds.

(b) Figure 5: fitted risk changes from return time series

Risk changes from time series as a function of January–September returns

7 Interpretive synthesis

- **The paper is about incentives, not just performance.** The fund manager’s problem is not whether to beat the market on average, but how to choose risk when investor flows reward particular year-end outcomes.
- **Investor learning creates nonlinear payoffs.** Young funds are evaluated more heavily on recent returns, producing a steeper and more nonlinear flow curve.

- **Risk incentives are state dependent.** A fund’s optimal risk manipulation depends on whether it is behind, near, moderately ahead, or far ahead of the market.
- **The evidence is strongest where theory predicts.** Young and small funds, where flows are most sensitive and portfolio changes are easier, respond most strongly.
- **The result is an agency distortion.** Investors may use performance as a sensible signal of ability, but the aggregate response creates a payoff function that encourages risk choices not necessarily aligned with investors’ risk-adjusted objectives.

8 Short summary

- The paper studies whether mutual funds change end-of-year risk in response to incentives created by the flow-performance relationship.
- The authors estimate the flow-performance relationship semi-parametrically using Morningstar data from 1983–1993.
- Young funds have a much steeper and more nonlinear flow-performance curve than old funds.
- The flow curve creates risk-taking incentives that depend on January–September performance.
- A fund slightly behind the market may want to gamble because catching up produces inflows while falling further behind has limited marginal cost.
- A fund moderately ahead may prefer to reduce risk and lock in gains.
- A fund far ahead may again want to gamble because extreme winners attract large flows.
- Holdings data show that funds change September-to-December risk in ways related to the computed incentives.
- The response is strongest among smaller funds.
- Detailed piecewise results show young funds moderately behind the market increasing risk, broadly matching the incentive curve.
- Return-based time-series risk checks support the broad pattern but are less precise.
- The main contribution is to treat investor flows as an implicit incentive contract and to show that funds respond to it.

9 Conclusion

9.1 Core conclusion

Chevalier and Ellison (1997) show that mutual fund companies respond to the implicit incentives created by investor flows. Because investors allocate capital based on performance in a nonlinear way, fund companies may change portfolio risk near year-end to increase expected future inflows.

9.2 Economic intuition

The key intuition is option-like. When a fund is moderately behind, increasing risk gives it a chance to recover and attract flows, while a further loss may not make flows much worse. When a fund is far ahead, additional upside may place it among star performers and generate large inflows. In both cases, the flow-performance curve rewards risk taking even if the risk is not optimal for investors.

9.3 Incremental lesson

The paper's lasting contribution is methodological and substantive. Methodologically, it maps an empirically estimated payoff function into managerial incentives. Substantively, it shows that mutual fund risk choices can be shaped by business incentives rather than by a pure objective of maximizing investor risk-adjusted returns.

9.4 Takeaway

Investor discipline through flows is imperfect. Flows reward past returns and can discipline poor managers, but their nonlinear shape also creates risk-taking distortions. The mutual fund market is therefore a setting where market incentives both discipline and distort delegated portfolio management.